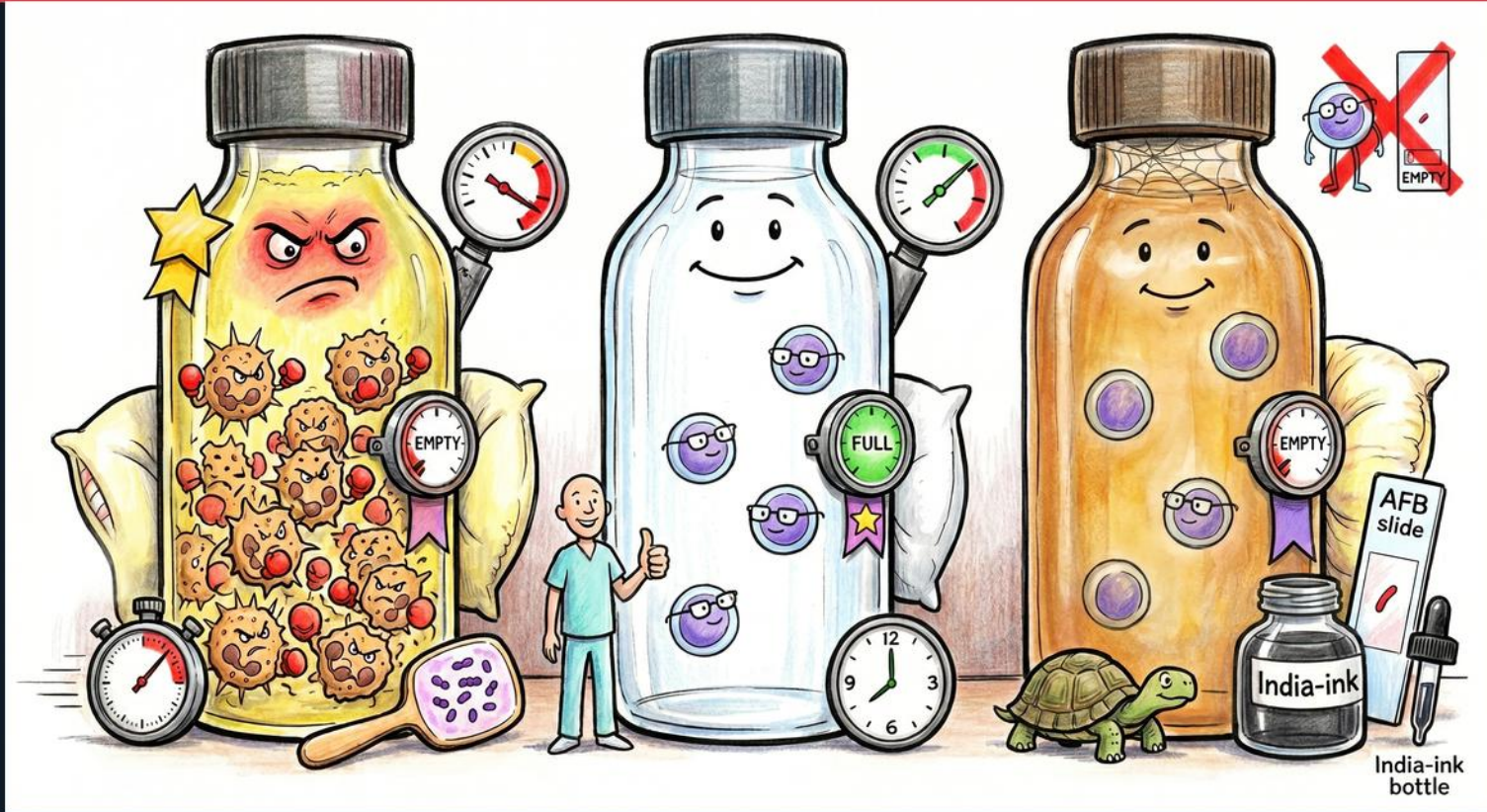


Meningitis — CSF Profiles (Bacterial vs Viral vs TB/Fungal)





Bacterial vial: angry, cloudy yellow

WHAT YOU SEE

Furious red-eyed vial of cloudy yellow fluid stuffed with cookie-like cells — bacterial CSF: turbid, packed with neutrophils.

WHAT IT ENCODES

Bacterial CSF is CLOUDY/turbid from purulent cellular exudate. WBC is high, typically 1,000-5,000 cells/uL (range 100-10,000) with NEUTROPHIL (PMN) predominance >80%. Gram stain positive in 60-90%, culture ~80% (yield falls after antibiotics). The full signature is high count + PMN + low glucose + high protein + high pressure.

THE BOARD TRAP

"any PMN-predominant CSF is bacterial" -> wrong; very EARLY viral meningitis (first 24-48h) can be PMN-predominant — glucose is the discriminator (viral keeps it normal). Repeat LP in 12-24h if unsure.



Bacterial: high opening pressure

WHAT YOU SEE

Pressure gauge pinned in the red zone beside the bacterial vial — markedly elevated opening pressure.

WHAT IT ENCODES

Bacterial meningitis raises opening pressure (>180 mmH₂O, often 200-500) in ~90% of cases, from purulent exudate blocking CSF resorption at arachnoid granulations. Normal OP is 6-20 cmH₂O (~80-200 mmH₂O), measured lateral decubitus with legs extended. Very high OP also flags cryptococcal meningitis (may need serial LPs).

THE BOARD TRAP

"normal opening pressure rules out bacterial" -> wrong; OP can be normal early or in partially treated cases — diagnosis rests on the full CSF profile, not pressure alone.



Bacterial: cookie cells = neutrophils

WHAT YOU SEE

Brown cookie-textured cells crammed inside the angry vial — the neutrophil-rich pleocytosis.

WHAT IT ENCODES

The packed cookie-cells represent the dense PMN pleocytosis of pyogenic bacterial meningitis. Neutrophil-dominant CSF with markedly LOW glucose is the bacterial hallmark. Empiric ceftriaxone + vancomycin (+ ampicillin if Listeria-risk) plus dexamethasone should not wait for confirmation.



Bacterial: low glucose, high protein

WHAT YOU SEE

The 'EMPTY' fuel-gauge tag on the bacterial vial — glucose drained low while protein runs high.

WHAT IT ENCODES

Bacterial CSF = LOW glucose (<40 mg/dL or ratio <0.4) and HIGH protein (often 100-200+; normal <45). Low glucose reflects bacterial/leukocyte metabolism and impaired transport; high protein reflects blood-brain-barrier breakdown.

THE BOARD TRAP

"protein only mildly elevated, so it's viral" -> wrong; partially treated bacterial meningitis blunts the classic findings — recent antibiotics are the clue, and empiric therapy should not be withheld.



Viral vial: clear & smiling

WHAT YOU SEE

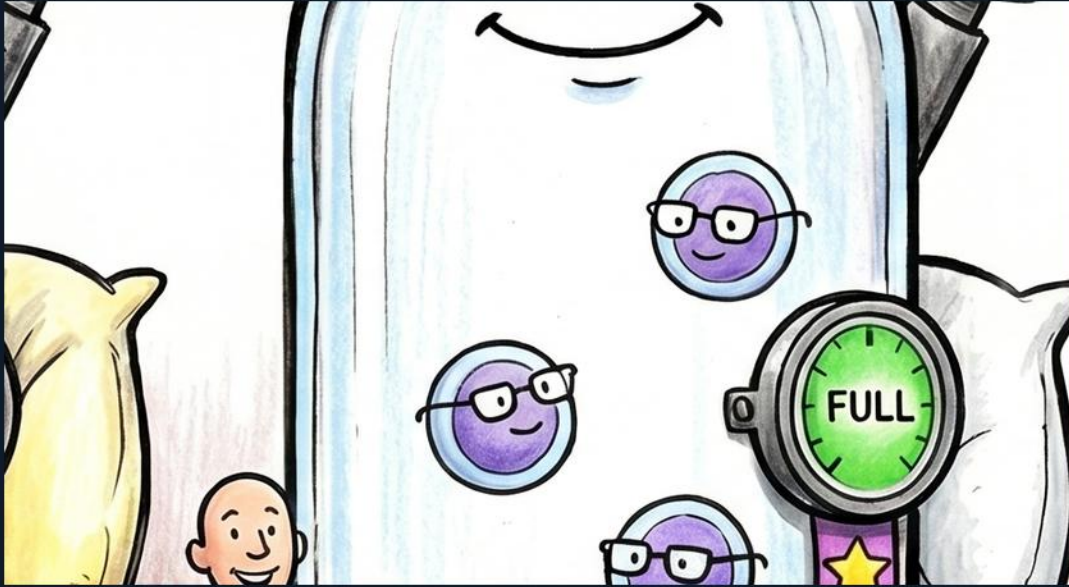
Calm, smiling vial of crystal-clear fluid in the center — viral (aseptic) CSF: clear and well-appearing.

WHAT IT ENCODES

Viral (aseptic) CSF is CLEAR/colorless with normal-to-mildly-elevated opening pressure. Enteroviruses cause >85% (summer-fall); diagnose by CSF RT-PCR. Patients are typically well with intact mentation — altered mentation redirects you to encephalitis or bacterial disease.

THE BOARD TRAP

"clear CSF means no meningitis" -> wrong; early bacterial or partially treated cases can also look grossly clear — cell differential and glucose define the category, not gross appearance.



Viral: lymphocytes (eyeglass cells)

WHAT YOU SEE

Round purple cells wearing eyeglasses inside the clear vial — mononuclear lymphocytes.

WHAT IT ENCODES

Viral CSF shows a modest LYMPHOCYTIC pleocytosis, usually 10-500 cells/uL (rarely >1,000), mononuclear-predominant — far lower than the thousands of bacterial disease. Enterovirus, HSV-2, VZV, and West Nile all produce this lymphocytic pattern.

THE BOARD TRAP

"first tap shows PMNs, so not viral" -> wrong; early enteroviral meningitis often has an initial neutrophilic shift converting to lymphocytic within 12-24h — preserved glucose distinguishes it.



Viral: normal glucose (FULL gauge)

WHAT YOU SEE

Green 'FULL' fuel-gauge tag and gold star on the clear vial — glucose preserved, the key viral discriminator.

WHAT IT ENCODES

Viral CSF = NORMAL glucose (ratio >0.4 - 0.6) with normal-to-mildly-elevated protein (20-80 mg/dL). NORMAL glucose is THE single best discriminator separating viral from bacterial/TB/fungal. Mumps and LCMV are rare viral exceptions that can mildly lower glucose.

THE BOARD TRAP

"lymphocytic CSF with LOW glucose = viral" -> wrong; low glucose plus lymphocytes is the hallmark of TB/fungal (or sometimes Listeria), NOT viral — viral preserves glucose.



Glucose:serum ratio (compute it)

WHAT YOU SEE

The thumbs-up clinician between the vials — the bedside reminder to compute the CSF:serum glucose ratio, not the raw number.

WHAT IT ENCODES

Always compute the CSF:serum glucose RATIO, not absolute CSF glucose. Ratio < 0.4 indicates a glucose-consuming process (bacterial, TB, fungal); normal is ~ 0.6 (CSF $\approx$ two-thirds of serum). Example: CSF 50 with serum 150 = 0.33, abnormal despite a 'normal-looking' 50.

THE BOARD TRAP

"CSF glucose 50 looks normal, call it viral" $\rightarrow$ wrong; a normal-looking absolute glucose can still be LOW relative to a high serum — the ratio (0.33) is the discriminator, especially in hyperglycemic/diabetic patients.



TB/Fungal vial: hazy & cobwebbed

WHAT YOU SEE

Amber, hazy vial draped in cobwebs on the right — TB/fungal CSF: xanthochromic, subacute, with a spider-web pellicle.

WHAT IT ENCODES

TB and fungal (cryptococcal, coccidioidal) meningitis produce HAZY/xanthochromic CSF over a subacute-to-chronic course of weeks. A pellicle (spider-web clot) may form on standing in TB. Cryptococcus is found by India ink (~70-80%) or, better, CSF cryptococcal antigen (CrAg, >95%), especially in HIV with CD4 <100.

THE BOARD TRAP

"chronic lymphocytic meningitis must be infectious" -> wrong; the chronic basilar lymphocytic differential also includes neurosyphilis, sarcoidosis, and carcinomatous meningitis — broaden the workup.



TB/Fungal: lymphocytes + low glucose

WHAT YOU SEE

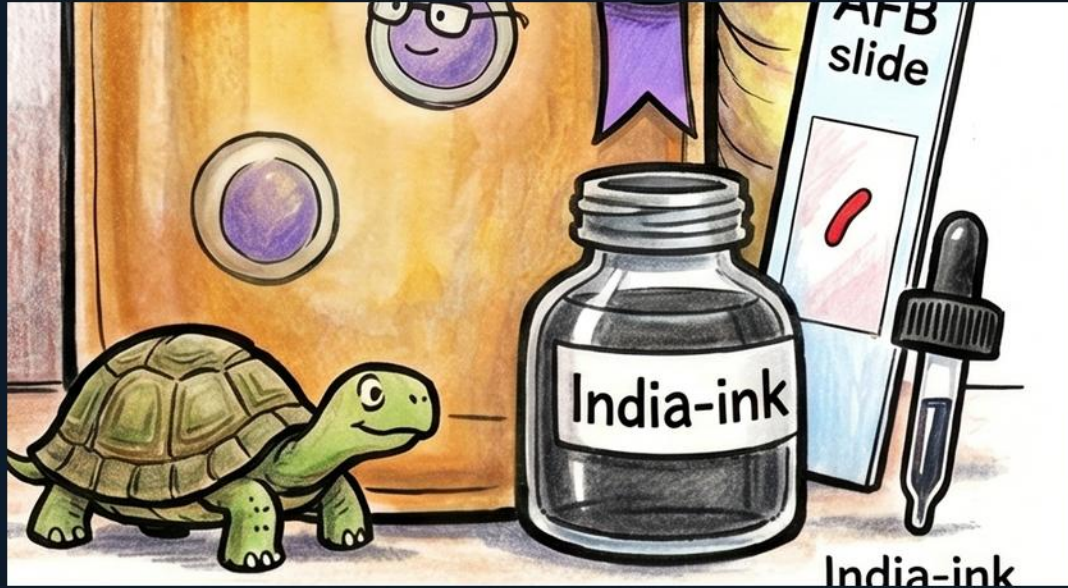
Purple eyeglass-cells inside the hazy vial with an 'EMPTY' glucose tag — lymphocytic pleocytosis with drained glucose.

WHAT IT ENCODES

TB/fungal CSF = LYMPHOCYTIC pleocytosis (typically 100-500/uL) with markedly LOW glucose and very HIGH protein (100-500 mg/dL; >1 g/dL in TB with CSF block) — biochemically mimicking bacterial but with a lymphocytic rather than neutrophilic differential. Early TB can have a transient PMN phase.

THE BOARD TRAP

"lymphocytic = viral, stop here" -> wrong; lymphocytes with LOW glucose and HIGH protein over weeks is TB/fungal until proven otherwise — send AFB smear/culture, Xpert MTB/RIF, and cryptococcal antigen.



India ink bottle (Cryptococcus)

WHAT YOU SEE

Labeled 'India-ink' bottle with dropper at the base of the hazy vial — the stain that reveals encapsulated *Cryptococcus*.

WHAT IT ENCODES

India ink shows the clear capsular halo of *Cryptococcus neoformans* (~70-80% sensitive), but CSF cryptococcal antigen (CrAg) is preferred (>95% sensitive/specific). Cryptococcal meningitis classically occurs in HIV with CD4 <100; high opening pressure is managed with serial therapeutic LPs.



AFB slide (low-yield TB smear)

WHAT YOU SEE

The 'AFB slide' card propped by the hazy vial — the acid-fast smear for TB, notoriously insensitive.

WHAT IT ENCODES

TB meningitis AFB smear yield is LOW (10-40%), so send LARGE-volume CSF (≥ 6 -10 mL, ideally serial taps) for AFB smear, mycobacterial culture (gold standard, slow 2-6wk), and NAAT (Xpert MTB/RIF Ultra). Start empiric RIPE plus adjunctive corticosteroids on clinical suspicion — do not wait for culture.

THE BOARD TRAP

"negative AFB smear rules out TB" -> wrong; a single negative smear can't exclude TB given poor sensitivity — need multiple large-volume taps, culture, and PCR, with empiric treatment when suspicion is high.



Tortoise / clock (subacute tempo)

WHAT YOU SEE

Slow tortoise and a wall clock at the foot of the hazy vial — the weeks-long subacute course pointing to TB/fungal.

WHAT IT ENCODES

Tempo discriminates: acute bacterial evolves over hours to 1-2 days, viral over days, TB/fungal SUBACUTELY over weeks. Weeks of headache, low-grade fever, and malaise with lymphocytic, low-glucose CSF should trigger a TB/fungal workup.

THE BOARD TRAP

"default to acute pyogenic therapy despite weeks of symptoms" -> wrong; the subacute timeline argues against typical pyogenic bacteria and for TB/fungal (or partially treated bacterial).



Crossed-out empty viral cells (early PMN trap)

WHAT YOU SEE

Red-X over the small 'EMPTY' eyeglass-cells in the top-right corner — the reminder that profiles can mislead early.

WHAT IT ENCODES

Master pattern: BACTERIAL = cloudy, PMN, glucose LOW, protein HIGH, OP high; VIRAL = clear, lymphocytes, glucose NORMAL; TB/FUNGAL = hazy, lymphocytes, glucose LOW, protein VERY high, subacute. The two-variable shortcut is CELL TYPE (PMN vs lymphocyte) × GLUCOSE (low vs normal), plus tempo and recent antibiotics. Normal CSF: WBC <5 (mononuclear), glucose ratio ~0.6, protein 15-45, OP 6-20 cmH₂O.

THE BOARD TRAP

"memorize cell type alone" -> wrong; the classic trap is lymphocytes-with-LOW-glucose (TB/fungal) mislabeled viral, or early-PMN viral mislabeled bacterial — read cell type AND glucose together.

Summary — every symbol

1. Bacterial vial: angry, cloudy yellow

Bacterial CSF is CLOUDY/turbid from purulent cellular exudate

2. Bacterial: high opening pressure

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3. Bacterial: cookie cells = neutrophils

The packed cookie-cells represent the dense PMN pleocytosis of pyogenic bacterial meningitis

4. Bacterial: low glucose, high protein

Bacterial CSF = LOW glucose (<40 mg/dL or ratio <0.4) and HIGH protein (often 100-200+; normal <45)

5. Viral vial: clear & smiling

Viral (aseptic) CSF is CLEAR/colorless with normal-to-mildly-elevated opening pressure

6. Viral: lymphocytes (eyeglass cells)

Viral CSF shows a modest LYMPHOCYTIC pleocytosis, usually 10-500 cells/uL (rarely $>1,000$), mononuclear-predominant — far lower than the thousands of bacterial disease

7. Viral: normal glucose (FULL gauge)

Viral CSF = NORMAL glucose (ratio >0.4 - 0.6) with normal-to-mildly-elevated protein (20-80 mg/dL)

8. Glucose:serum ratio (compute it)

Always compute the CSF:serum glucose RATIO, not absolute CSF glucose

9. TB/Fungal vial: hazy & cobwebbed

TB and fungal (cryptococcal, coccidioidal) meningitis produce HAZY/xanthochromic CSF over a subacute-to-chronic course of weeks

10. TB/Fungal: lymphocytes + low glucose

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11. India ink bottle (Cryptococcus)

India ink shows the clear capsular halo of *Cryptococcus neoformans* (~70-80% sensitive), but CSF cryptococcal antigen (CrAg) is preferred ($>95\%$ sensitive/specific)

12. AFB slide (low-yield TB smear)

TB meningitis AFB smear yield is LOW (10-40%), so send LARGE-volume CSF (≥ 10 mL, ideally serial taps) for AFB smear, mycobacterial culture (gold standard, slow 2-6wk), and NAAT (Xpert MTB/RIF Ultra)

13. Tortoise / clock (subacute tempo)

Tempo discriminates: acute bacterial evolves over hours to 1-2 days, viral over days, TB/fungal SUBACUTELY over weeks

14. Crossed-out empty viral cells (early PMN trap)

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